AI-Powered Early Diabetes Detection

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Abstract

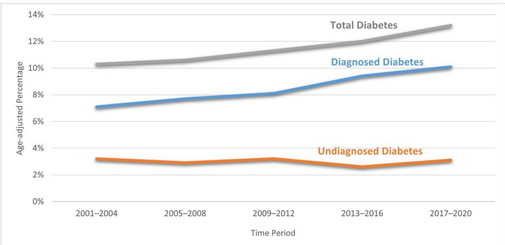
Diabetes is a chronic disease that affects millions of Americans.

- Approximately 11.6% of the U.S population has diabetes.
- 22.8% of adults with diabetes are undiagnosed (Statistics About Diabetes | ADA, n.d.).

Machine learning enhances healthcare efficiency and patient outcomes by enabling early diabetes detection from clinical and lifestyle factors.

A diabetes dataset was analyzed using ML methods, including logistic regression, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM). Training and validation data optimized model parameters, while test data evaluated final performance. Naive Bayes outperformed other models in predicting prediabetes and diabetes, based on F1 score and Recall. This study adds to the literature on using ML for disease prediction, with further analysis incorporating additional variables as a potential next step.

Trends in Prevalence of Diagnosed Diabetes, Undiagnosed Diabetes, and Total Diabetes Among Adults Aged 18 Years or Older, United States, 2001–2004 to 2017–2020



Introduction & Significance

About Diabetes

Diabetes is a chronic disease where the pancreas cannot produce enough insulin hormone to regulate blood sugar levels to stay within healthy limits. Diabetes symptoms include increased thirst, frequent urination and can cause heart disease, stroke, and kidney failure without diagnosis and treatment.

In 2021, around 11% of cardiovascular deaths were caused by high blood glucose levels. Diabetes is often diagnosed through invasive blood testing. There is a global target to stop the rise in diabetes by 2025 (World Health Organization: WHO & World Health Organization: WHO, 2024).

Diabetes Types

- **Type I**: Typically diagnosed in childhood. The immune system attacks and destroys insulin-producing cells.
- **Type II:** Develops in adulthood as a result of a high-glycemic diet and lifestyle factors. This type is more common, yet symptoms can be mild and take years to notice, often after complications develop.

Diabetes in Machine Learning Literature

Early detection is important to prevent more serious complications. Machine learning methods can improve healthcare professionals' ability to catch diabetes early. This enhances diagnostic monitoring and patient lives.

A study by Iparraguirre-Villanueva et al. (2023) uses KNN, Bernoulli Naive Bayes (BNB), decision trees, logistic regression and SVM models to classify and detect type II diabetes. The study analyzed whether the number of pregnancies, blood glucose concentration, BMI, and more contributed to classifying diabetes in Pima American Indians. KNN and BNB using Synthetic Minority Over-Sampling Technique (SMOTE) for class imbalance were the best performing models.

Other studies such as Turi et al. (2017) used a Multivariate Adaptive Regression Splines (MARS) model to predict diabetes using various interactions between the data to understand collective effects.

Methods

Data

For this analysis, we use data from the UC Irvine Machine Learning Repository, which originally sourced the data from the CDC. There are 253,680 observations total and 21 independent variables. Our target variable is predicting whether an individual will develop diabetes.

Example variables:

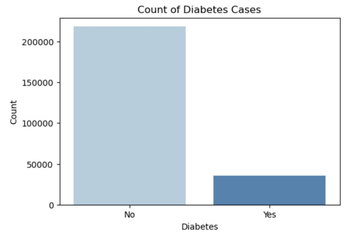
- *HighBP* : 0 = normal BP levels, 1 = high BP
- *HighChol*: 0 = normal cholesterol levels, 1 = high cholesterol
- *CholCheck*: 0 = no cholesterol check in 5 years, 1 = cholesterol check in 5 years
- *BMI* : Body Mass Index

Exploratory Data Analysis (EDA)

For the EDA, we started by looking at summary statistics for each variable, using box plots to spot outliers and histograms to understand feature distributions.

We generated a correlation matrix to explore relationships between features and the target variable (diabetes_binary). This helped identify which features were strongly related to diabetes and flagged potential multicollinearity, which we needed to address for models sensitive to it, such as logistic regression and SVM.

The target variable was found to be unbalanced, which can cause issues during model training. This was addressed by implementing SMOTE.



Data Preparation

In the data preparation stage, we performed feature engineering by adding six features from previous studies and those relevant to diabetes development. Some of these features included:

- *Age * Sex*
- *MentHealth * HighBP*
- *Education * HvyAlcohol*

These additions aim to improve the model's ability to identify key factors contributing to diabetes. Next, we split the data into 60% training, 20% validation, and 20% test sets. To address the class imbalance, we applied SMOTE - a method that generates new instances by oversampling the minority class, to the training data.

We applied MinMaxScaler to scale the data, ensuring consistency for distance-based models like KNN, SVM, and logistic regression.
These steps should enhance model performance in predicting diabetes. Lastly, LASSO regularization was performed for feature selection to help address multicollinearity concerns in certain models, but due to the size of the dataset, LASSO kept all variables.

Model Training

The models were fitted first on training data and then ran on validation data to make diabetes case predictions. This allowed us to assess and optimize hyperparameters without data leakage.

After hyper-tuning the models, they were tested and compared between each other based on the F1 score and recall. The ML models include:

- Logistic Regression
- LDA (Linear Discriminant Analysis)
- QDA (Quadratic Discriminant Analysis)
- KNN (K-Nearest Neighbors)
- Naïve Bayes
- SVM (Support Vector Machine)
- Decision Trees
- Bagging / Boosting
- Random Forest

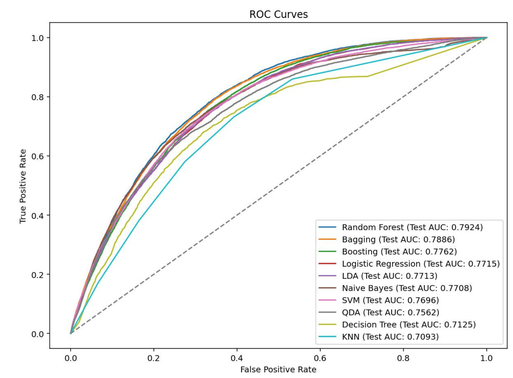
Results

Model Evaluation

The models were optimized for the recall values. Recall is a key metric in healthcare case studies, especially around disease prediction, where it is better to correctly classify many true positive cases, even at the cost of some false positives. This is so a diabetes case isn't undiagnosed, leading to complications. The F1 score is another important success metric – a harmonic balance between both precision and recall.

- Recall values across all models ranged from 51% to 74%
- F1 scores across all models ranged from 42% to 80%

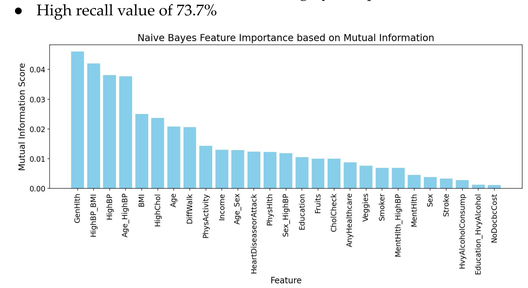
The ROC curve compares the tradeoff between the true positive rate (TPR) and false positive rate (FPR) results of the models at different classification thresholds.



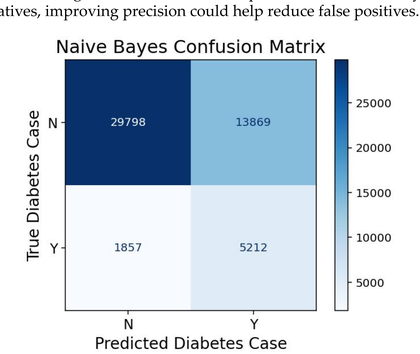
Final Model - Naïve Bayes

The final model selected was Naïve Bayes. This model is the best performing model for this use case in many ways:

- One of the top performing models shown in the ROC Curves
- Given feature likelihood, it remains highly interpretable
- High recall value of 73.7%



The Naïve Bayes model achieved 25% precision, 74% recall (or TPR), a 74% F1 score, and 70% accuracy. The model correctly identified 29,798 true negatives and 5,212 true positives but misclassified 32% as false positives and 26% as false negatives. While the model performs well in identifying true negatives, improving precision could help reduce false positives.



Discussion

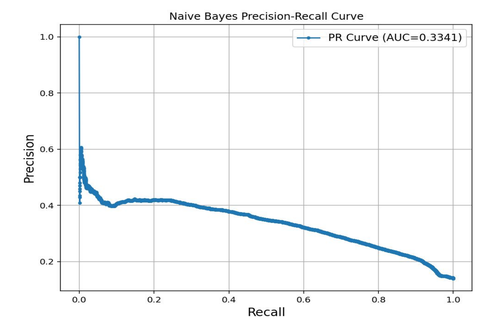
Key Takeaways

- Due to the low dimensional data, LASSO was not particularly effective
- SMOTE assisted in the training process to achieve better performance
- Hyperparameter tuning for recall lead to strong results
- Simpler models performed similarly to more complex, if not better
- Features like GenHlth (GeneralHealth) and HighBP_BMI contributed most

Further Areas of Investigation

- Genetic Variables
- Demographic Variables
  - Family Life/History
  - Employment
  - Location
- Blood Glucose Level Variable

Our study provided insights into various personal and lifestyle factors contributing to type II diabetes. However, a deeper understanding of all influencing factors would be valuable. Expanding the dataset could improve the generalizability of predictions, while incorporating timestamps for these factors could offer a clearer picture of when diabetes risk becomes significant in a patient's life.



Conclusion

In conclusion, the best model for this use case was determined to be Naïve Bayes, based on the recall value of 74%, F1 score of 74% and the model interpretability.

This study provides analysis on how ML can help with early detection of diabetes, making diabetes diagnostics and management more efficient for healthcare providers.

With early monitoring of diabetes detection, doctors can conduct more targeted and purposeful blood tests, especially for patients who need it most.

References

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